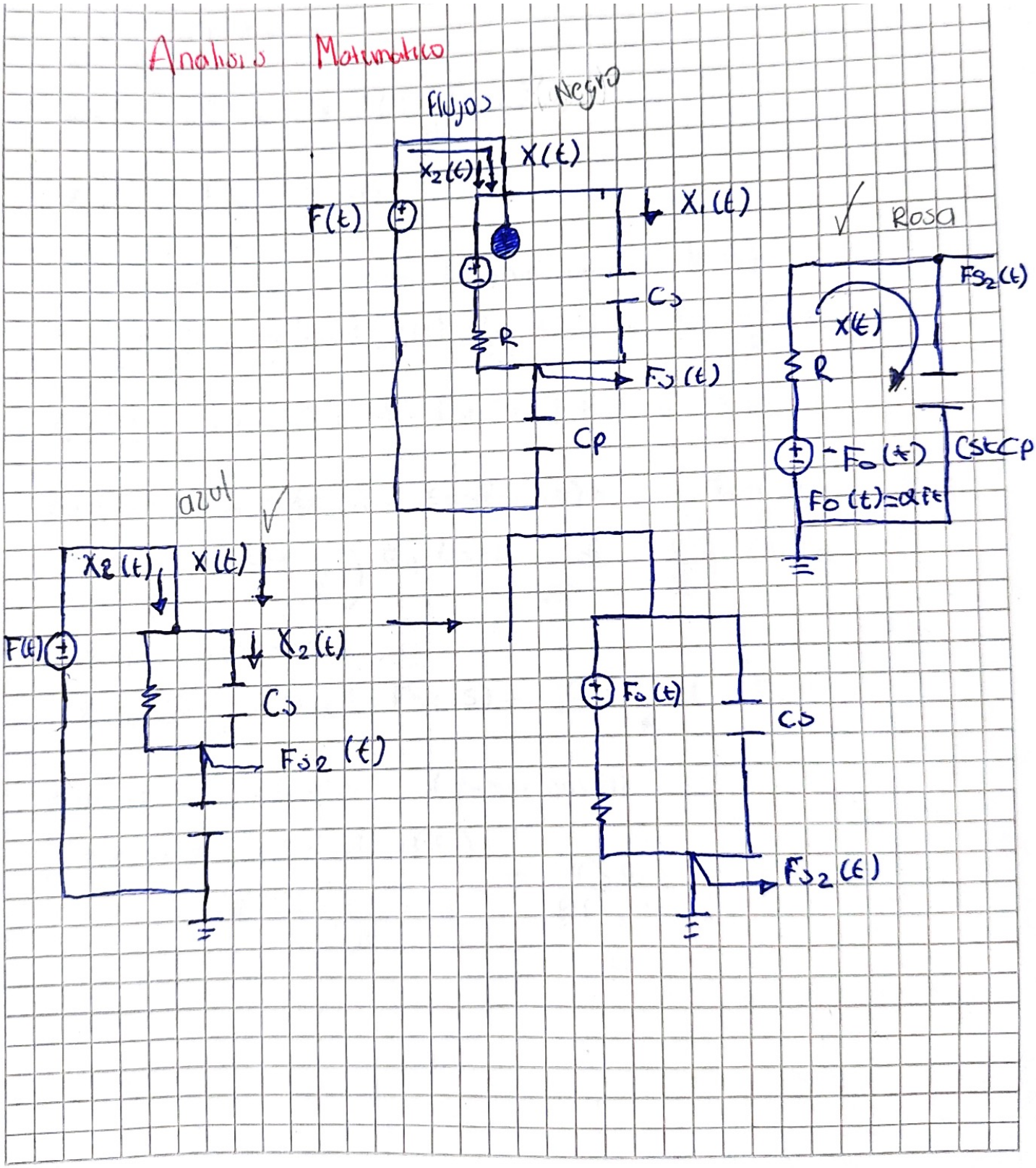
Analisis Matematico



Analisis del circuito por superposición

Circuito 1: Apagando $F_o(t)$ — Flujos

$X(t) = X_1(t) + X_2(t)$

$X(t) =$

$$X_1(t) = C_s \frac{d[F_s F(t) - F_{s1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

Relacionar
Transformada de Laplace (multiplicadores de S)

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p S F_{s1}(s) = C_s S [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p S F_{s1}(s) = C_s S F(s) - C_s S F_{s1}(s) + \frac{F_s}{R} - \frac{F_{s1}(s)}{R}$$

$$\left(C_p S + C_s S + \frac{1}{R}\right) F_{s1}(s) = \left(C_s S + \frac{1}{R}\right) F_s$$

Suma algebraica

$$\frac{R(C_p + C_s)S + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

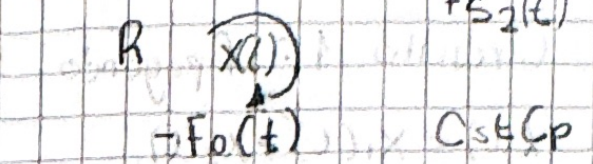
Calcular la salida (Primer salida)

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R} F(s)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

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$$F_s(t) = F_{s1}(t) + F_{s2}(t)$$

Circuito 2: apagando $F(t)$



$$X(t) = X_2(t)$$

$$X_1(t) = C_s + C_p \left[\frac{F_{s1}(t)}{dt} \right]$$

$$F_o(t) = \alpha F(t)$$

$$X_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$-\alpha F(t) = R x(t) + \frac{1}{C_p + C_s} \int X(t)\, dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s} \int X(t)\, dt$$

Transformada de Laplace

$$-\alpha F(s) = R X(s) + \frac{X(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{X(s)}{(C_p + C_s)s}$$

para 8 calcular la salida

$$F(s) = -\frac{X(s)}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}$$

$$\frac{F_{s2}(s)}{F(s)} = \frac{\frac{X_s}{(C_p + C_s)s}}{-\frac{X(s)}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}} = -\frac{\alpha}{RC(C_p + C_s)s + 1}$$

$$F_{s2}(s) = -\frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

Función de transferencia

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s) + 1}$$